

Lecture 18

STAT 109: Introductory Biostatistics

Lecture 18: Sampling Distributions and Estimation

Learning outcomes

- Define the “sampling distribution” of a statistic.
 - Describe confidence level, parameter, and upper and lower bounds of an interval in the context of an estimation problem.
 - Compute a confidence interval for a population proportion.
 - Interpret a confidence interval for a population proportion.
 - Compute a confidence interval for a population mean.
 - Interpret a confidence interval for a population mean.
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Why estimation?

Descriptive statistics summarize the information in a dataset. The goal of **inferential statistics** is to infer something about a **population** from a **sample**.

Definitions: - **Population vs sample:** We observe information from a subset of people or objects: this set is the **sample**; in inferential statistics, we want to say something about the population (e.g. all people or objects in our research question), not just the ones in our sample. - **Parameter vs statistic:** A **parameter** is a number describing the population: usually this is unknown as we only have information from a subset of the population (a sample); a **statistic** is a number computed from the sample (e.g. sample proportion, sample mean).

Example: Is Moderna’s vaccine effective?

[Moderna vaccine trial \(Reuters\)](#)

What is the population in this context? The sample? What is the parameter and what is the statistic?

Example setting: St. Joe’s in Eureka, California

St. Joe’s Hospital in Eureka, California performs a certain **heart procedure**. We are interested in whether the observed mortality rate at St. Joe’s for this procedure is higher than the national rate.

National mortality rate (benchmark): $p_0 = 0.08$ (8%).

Two sets of data:

- **Large sample:** 200 patients, with 14 deaths ($x = 14$, $n = 200$).
- **Small sample:** 16 patients, with 3 deaths ($x = 3$, $n = 16$).
- **Population:** All patients who could receive this heart procedure (or all such patients at similar hospitals).
- **Parameter:** The true mortality rate (proportion) at St. Joe's, often denoted π or p .
- **Statistic:** The proportion who died in the sample: $\hat{p} = x/n$.

General notation and sampling distribution (proportion)

For a binary outcome (e.g. died vs. survived):

- **Parameter:** population proportion π (or p).
- **Statistic:** sample proportion $\hat{p} = (\text{number of successes}) / n$.

The **sampling distribution** of $\hat{p}$ can be:

- Simulated (e.g. using a for loop and mean function in R with a large number of replicates),
- Computed exactly using the binomial distribution (e.g. dbinom/pbinom in R),
- Approximated with the normal distribution when n is large enough (e.g. np and n(1-p) are both at least 10).

Use the [One Proportion applet](#) to compare the observed sample proportion to the sampling distribution if St. Joe's had the same mortality rate as the national rate (8%).

General formula for a 95% confidence interval

For a **95% confidence interval**, we use the “2 standard deviations” rule from the Empirical Rule (approximate):

Statistic $\pm 2 \times$ (standard deviation of statistic)

The “standard deviation of the statistic” is also called the **standard error (SE)**. So:

95% CI statistic $\pm 2 \times$ SE

Two important cases:

Formula 1: One proportion (Wald formula)

For a **population proportion** π , we estimate it with the sample proportion $\hat{p}$. The standard error of $\hat{p}$ is:

$$SE(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

So the **95% confidence interval for a single proportion** is:

$$\hat{p} \pm 2\sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Conditions for Formula 1 (Wald):

1. **Binomial random process:** (a) **Binary outcome** (success/failure), (b) **Independent trials**, (c) **Number of trials** n fixed in advance, (d) **Same probability of success** π on each trial.
 2. **Normal approximation:** $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$ (at least 10 successes and 10 failures in the sample, so the sampling distribution of $\hat{p}$ is approximately normal).
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Formula 2: One mean

For a **population mean** μ , we estimate it with the sample mean $\bar{x}$. The standard error of $\bar{x}$ is:

$$SE(\bar{x}) = \frac{s}{\sqrt{n}}$$

where s is the sample standard deviation. So the **95% confidence interval for a single mean** is:

$$\bar{x} \pm 2 \frac{s}{\sqrt{n}}$$

(For small samples, we often use the t distribution so the multiplier is slightly larger than 2; here we use 2 for a simple “by hand” rule.)

Conditions for Formula 2:

1. **Either** the population is (approximately) **normal**, **or** the sample size is $n > 30$ (so the sampling distribution of $\bar{x}$ is approximately normal by the Central Limit Theorem).
 2. The sample is a random sample (or can be treated as such) from the population.
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Example 1: Confidence interval for a proportion (St. Joe’s) — large sample

Goal: Estimate the mortality rate at St. Joe’s (Eureka, CA) for this heart procedure using the **large sample** (200 patients, 14 deaths), with **95% confidence**. The national benchmark is $p_0 = 0.08$ (8%).

- **Data:** $n = 200$, $x = 14$ deaths.
- **Statistic:** $\hat{p} = 14/200 = 0.07$ (7%).
- **Standard error:**

$$SE(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{\frac{0.07 \times 0.93}{200}} \approx \sqrt{0.0003255} \approx 0.0180.$$

- **95% CI:** $\hat{p} \pm 2 \times SE(\hat{p}) = 0.07 \pm 2(0.0180) = 0.07 \pm 0.036$.
- **Interval:** Lower bound ≈ 0.034 , upper bound ≈ 0.106 , so about **[0.034, 0.106]** or **[3.4%, 10.6%]**.

Interpretation: We are 95% confident that the true mortality rate for this heart procedure at St. Joe’s is between about 3.4% and 10.6%. The national rate (8%) lies inside this interval, so the data are consistent with St. Joe’s having the same mortality rate as the national benchmark.

Small sample for a proportion: Plus Four

When the sample size is small, the condition for the normal approximation may not be satisfied. Then:

- For a **95% confidence interval** for a proportion, use the **Plus Four** formula.
- If the confidence level is not 95% and there are at least 10 successes and 10 failures in the sample, you may use the Wald (z) formula. Otherwise, consult a statistician.

Plus Four formula: Add 2 “successes” and 2 “failures” to the sample. With x = number of successes and n = sample size, define

$$\tilde{p} = \frac{x + 2}{n + 4}$$

and use the **effective sample size** $n + 4$. The standard error of $\tilde{p}$ is

$$SE(\tilde{p}) = \sqrt{\frac{\tilde{p}(1-\tilde{p})}{n+4}}.$$

The **95% confidence interval for a proportion (Plus Four)** is:

$$\tilde{p} \pm 2\sqrt{\frac{\tilde{p}(1-\tilde{p})}{n+4}}.$$

Example (small sample): Estimate the mortality rate at St. Joe's using the **small sample (16 patients, 3 deaths)** with 95% confidence. Use the Plus Four method.

- **Data:** $x = 3$ deaths (successes), $n = 16$ patients.
- **Plus Four estimate:** $\tilde{p} = \frac{x+2}{n+4} = \frac{3+2}{16+4} = \frac{5}{20} = 0.25$.
- **Standard error:** $SE(\tilde{p}) = \sqrt{\frac{\tilde{p}(1-\tilde{p})}{n+4}} = \sqrt{\frac{0.25 \times 0.75}{20}} = \sqrt{0.009375} \approx 0.0968$.
- **95% CI:** $\tilde{p} \pm 2 \times SE(\tilde{p}) = 0.25 \pm 2(0.0968) = 0.25 \pm 0.194$.
- **Interval:** Lower bound ≈ 0.056 , upper bound ≈ 0.44 , so about **[0.06, 0.44]** or **[6%, 44%]**.

Interpretation: We are 95% confident that the true mortality rate for this heart procedure at St. Joe's is between about 6% and 44%. With only 16 patients, the interval is wide and the estimate is uncertain; the Plus Four method gives a plausible range that is more reliable than the Wald interval for such a small sample.

Example 2: Confidence interval for a mean (by hand)

Goal: Estimate the mean recovery time (in days) for patients after the advanced heart procedure using a small sample, with **95% confidence**.

Data (recovery time in days for $n = 5$ patients):

5, 7, 7, 9, 12

By hand:

- **Sample mean:** $\bar{x} = \frac{5+7+7+9+12}{5} = \frac{40}{5} = 8$ days.
- **Sample standard deviation:**
 Deviations from mean: $-3, -1, -1, 1, 4$.
 Squared deviations: $9, 1, 1, 1, 16$. Sum = 28.
 $s^2 = 28/(5-1) = 7$, so $s = \sqrt{7} \approx 2.65$ days.
- **Standard error:** $SE(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{2.65}{\sqrt{5}} \approx 1.18$ days.
- **95% CI:** $\bar{x} \pm 2 \times SE(\bar{x}) = 8 \pm 2(1.18) = 8 \pm 2.36$ days.
- **Interval:** Lower bound ≈ 5.64 days, upper bound ≈ 10.36 days, so about **[5.6, 10.4]** days.

Interpretation: We are 95% confident that the true mean recovery time (in days) for patients after this advanced heart procedure is between about 5.6 and 10.4 days. The 95% confidence means that if we were to repeat the data collection many times, 95% of the time the true mean recovery time would be within the intervals (e.g. lower and upper bounds) computed from the samples. This interval is based on only 5 patients, so it is wide and uncertain; a larger sample would give a narrower and more precise interval.